Magnetic susceptibility $\vec{M} = \chi_m \vec{H}$

Linear media $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \underbrace{\mu_0(1+\chi_m)}\vec{H}$

$\rightarrow \mu = \mu_0(1+\chi_m)$

permiability

<u>P6.15</u> Magnetostatic potential.

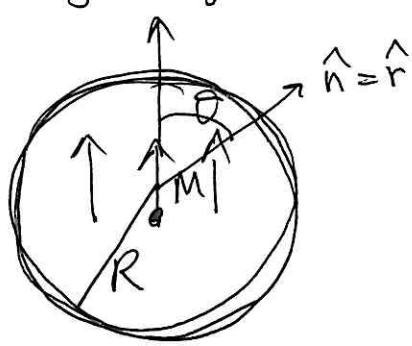
Problems with $\vec{J}_f = 0$ everywhere

$\Rightarrow \vec{\nabla} \times \vec{H} = 0 \Rightarrow$ can introduce $\vec{H} = -\vec{\nabla} W$

$\Rightarrow \vec{\nabla} \cdot \vec{H} + \vec{\nabla} \cdot \vec{M} = \vec{\nabla} \cdot \vec{B} = 0$

$\Rightarrow \nabla^2 W = +\underbrace{(\vec{\nabla} \cdot \vec{M})}_{\text{source of } W}$ $\Leftarrow$ Laplace equation with sources

$\Rightarrow$ Uniformly magnetized sphere $\vec{M} = M\hat{z}$



$\vec{\nabla} \cdot \vec{M} = 0$ everywhere but the surface

Surface "charge" $\vec{M} \cdot \hat{n} =$
$= M\hat{z} \cdot \hat{r} = M\cos\theta$

$\nabla^2 W = 0$ for $\begin{cases} r < R \\ r > R \end{cases}$

Eq. 3.65 $W(r,\theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}} \right) P_\ell(\cos\theta)$

$r<R \Rightarrow B_\ell = 0$ because $\frac{1}{r^{\ell+1}} \xrightarrow[r\to 0]{} \infty$ $\quad W_{in} = \sum A_\ell r^\ell P_\ell(\cos\theta)$

$r>R \Rightarrow A_\ell = 0$ because $r^{\ell+1} \xrightarrow[r\to\infty]{} \infty$ $\quad W_{out} = \sum \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos\theta)$

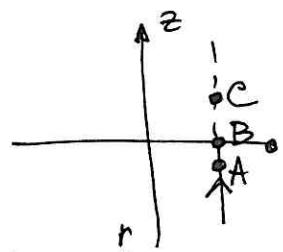
<u>BC</u> $H^{\parallel}_{out} = H^{\parallel}_{in}$ because $\vec{J}_f = 0$.

$H^{\perp}_{out} - H^{\perp}_{in} = M^{\perp}_{in}$

$W = -\int \vec{H}\cdot d\vec{\ell} \Rightarrow$ even for discontinuous $\vec{H}$ the integral is continuous

But $H^{\perp} = H_r = -\frac{\partial W}{\partial r}\hat{r}$

E.g. $\vec{H} = H\hat{z} \begin{cases} 0, & z>0 \\ 1, & z<0 \end{cases}$



$W(r) = -\int_{-\infty}^{r} H\,dz \Rightarrow \begin{cases} -H(r_A-) \text{ at } A \\ -H(r_B-) \text{ at } B \\ -H(r_B-) \text{ at } C \end{cases}$

since $\int_B^C 0\,dz = 0$.

$$\begin{cases} (2) & -\frac{\partial W_{out}(R,\theta)}{\partial r} + \frac{\partial W_{in}(R,\theta)}{\partial r} = M_r = M\cos\theta \\ (1) & W_{out}(R,\theta) = W_{in}(R,\theta) \end{cases}$$

$(1) \Rightarrow \sum_\ell A_\ell R^\ell P_\ell(\cos\theta) =$

$= \sum_\ell B_\ell \frac{1}{R^{\ell+1}} P_\ell(\cos\theta) \Rightarrow A_\ell = \frac{B_\ell}{R^{2\ell+1}} \Rightarrow B_\ell = A_\ell R^{2\ell+1}$.

(2) $\sum_\ell \frac{(\ell+1)B_\ell}{R^{\ell+2}} P_\ell(\cos\theta) + \sum_\ell \ell A_\ell R^{\ell-1} P_\ell(\cos\theta) = M\cos\theta$

$(2\ell+1) A_\ell P_\ell(\cos\theta) R^{\ell-1} = M\cos\theta$

$\begin{cases} P_0 = 1 \\ P_1 = x \\ P_2 = \frac{1}{2}(3x^2-1) \\ \vdots \end{cases}$

$\Rightarrow$ only $\ell=1$ works here: $A_\ell = 0$ for $\ell \neq 1$

$P_{\ell=1}(\cos\theta) = \cos\theta$: $\quad A_1 = \frac{1}{3} M$

$$\Rightarrow \quad W_m(r,\theta) = A_1 r\cos\theta = \frac{1}{3} M \underbrace{r\cos\theta}_{z} = \frac{1}{3} M z$$

$$\Rightarrow \quad \vec{H}_m(r,\theta) = -\vec{\nabla} W_m = -\frac{\partial W_m}{\partial r}\hat{r} - \frac{1}{r}\frac{\partial W}{\partial \theta}\hat{\theta} =$$

$$= -\frac{1}{3} M\cos\theta\,\hat{r} - \frac{1}{r}\left(\frac{1}{3} M r\right)(-\sin\theta)\hat{\theta} =$$

$$= -\frac{1}{3} M \underbrace{\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right)}_{\hat{z}} = -\frac{1}{3} M \hat{z}$$

Same is very clear in cartesian coordinates:

$$W = \frac{1}{3} M z \Rightarrow \vec{H}_m(x,y,z) = \left(-\frac{\partial}{\partial x}\hat{x} - \frac{\partial}{\partial y}\hat{y} - \frac{\partial}{\partial z}\hat{z}\right) W$$

$$= -\frac{1}{3} M \hat{z}.$$

$$\Rightarrow \quad \vec{B}_m = \mu_0 \left(\underbrace{\vec{H}}_{-\frac{1}{3}\vec{M}} + \vec{M}\right) = \frac{2}{3}\mu_0 \vec{M}.$$

* For the electric field of uniformly polarized sphere $\vec{E} = -\frac{1}{3\varepsilon_0}\vec{P}$ $\quad \left(\vec{H} = -\frac{1}{3}\vec{M}\right)$

<u>Ex. 4.2</u> $\quad \vec{D} = \varepsilon_0 \vec{E} + \vec{P} = \frac{2}{3}\vec{P} \quad \left(\vec{B} = \frac{2}{3}\mu_0 \vec{M}\right)$

$$W_{out}(r,\theta) = \frac{A_1 R^3}{r^2} P_1(\cos\theta) = \frac{M R^3 \cos\theta}{3 r^2}$$

$$\vec{H}_{out} = \frac{MR^3}{3}\left(-\cos\theta \frac{\partial}{\partial r}\left(\frac{1}{r^2}\right)\hat{r} - \frac{1}{r}\frac{1}{r^2}\left(\frac{\partial}{\partial\theta}\cos\theta\right)\hat{\theta}\right)$$

$$\cancel{= \frac{MR^3}{3r^3}\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right) = -\frac{MR^3}{3r^3}\hat{z}}$$

$$\vec{H}_{out} = + \frac{MR^3}{3} \left(\frac{2\cos\theta\,\hat{r}}{r^3} + \frac{\sin\theta\,\hat{\theta}}{r^3} \right) =$$

$$= \frac{MR^3}{3r^3} \left(3\cos\theta\,\hat{r} - \hat{z} \right) = \frac{MR^3}{3r^3} \left[3(\hat{m}\cdot\hat{r})\,\hat{r} - \hat{m} \right]$$

$$\hat{m} = \frac{\vec{M}}{M}$$

Use $\quad 2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta} = -\hat{z} + 3\cos\theta\,\hat{r}$

$$\hat{z} = \cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}$$

$$\cos\theta = \hat{z}\cdot\hat{r}$$

$$\vec{B}_{out} = \mu_0 \left(\vec{H}_{out} + \underset{\substack{\parallel \\ 0}}{\vec{M}_{out}} \right) = \frac{\mu_0 MR^3}{3r^3} \left[3(\hat{m}\cdot\hat{r})\,\hat{r} - \hat{m} \right]$$

$$= \frac{\mu_0 (m)}{4\pi r^3} \left[3(\hat{m}\cdot\hat{r})\,\hat{r} - \hat{m} \right]$$

where m = magnetic dipole moment of the sphere

$$= \frac{4\pi}{3} M R^3$$

↳ magnetization of unit volume

$$\vec{B}_m = \begin{cases} \frac{2}{3}\mu_0 \vec{M} & \text{(uniform)} \\ \frac{\mu_0 (m)}{4\pi r^3} \left[3(\hat{m}\cdot\hat{r})\,\hat{r} - \hat{m} \right] & \text{mag. field of a dipole of magnitude } \frac{4\pi}{3} R^3 M. \end{cases}$$